

Physical Security Evaluation of ASCON-128: Deep-Learning Side-Channel Analysis and Clock-Glitch Fault Injection

Prajwal R • Prajwal G S Basavaraj • Sachin S • Nitesh
Dept. of Electronics and Communication Engineering, PES University, Bengaluru, India
VLSID 2027 — Student Research Forum

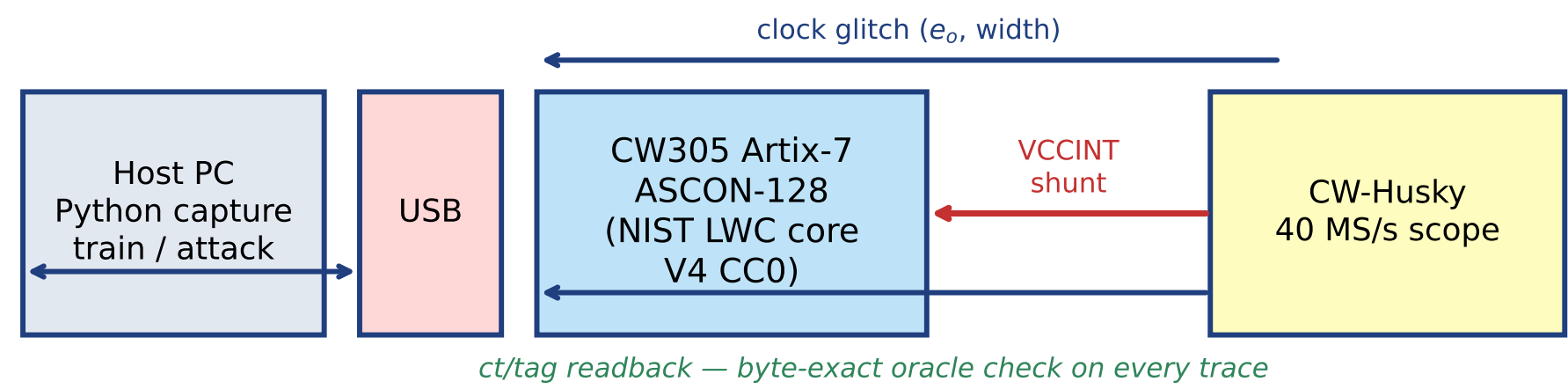
Contributions

- First hardware-measured DL-SCA + clock-glitch FI on the official unmasked NIST LWC ASCON-128 core (ascon-verilog V4) on CW305/CW-Husky — prior FPGA work attacked custom cores or software implementations only.
- Per-column key-bit recovery demonstrated: 42.7% mean key-rank top-1 vs 25% chance across all 64 S-box columns; closed-loop adaptive recovery at posterior 0.999 in 32 queries.
- Five leakage-validation gates separating genuine side-channel signal from five classes of confident false positives — two artifact classes unreported for LWC hardware.

Why ASCON-128 on Real Hardware?

ASCON-128 is the NIST SP 800-232 lightweight cryptography standard, entering constrained FPGA deployments. Its security under physical adversaries — power analysis and fault injection — determines whether the standard’s promise holds in the field. Published FPGA attacks evaluate custom ASCON cores (SCARL) or software implementations (ACNS’24 ensemble) — never the official unmasked LWC core. We measure that core directly.

Oracle-Verified Testbed

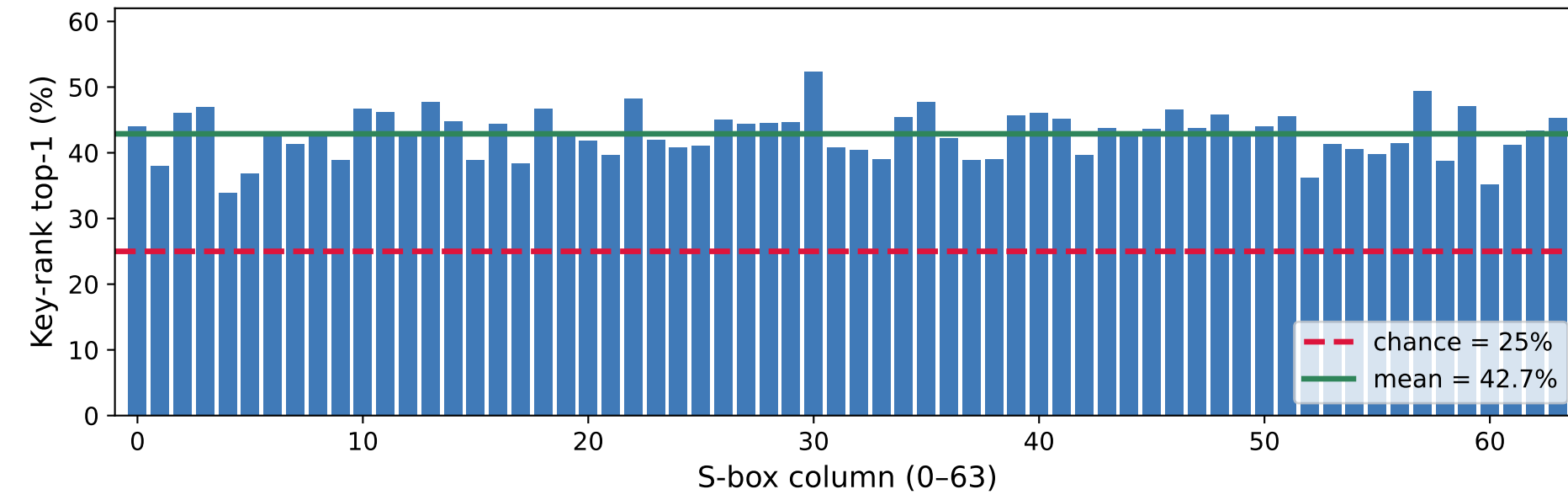


- CW305 Artix-7 + CW-Husky, 40MS/s vs 10MHz crypto clock (4 samples/cycle), VCCINT shunt, per-query trigger.
- Byte-exact oracle gate: every stored trace accepted only when its ct/tag readback matches a software reference (5/5 NIST KATs). Kills misconfigured-target errors before they poison datasets.
- Masked (d=1, 2-share) reference core on the same platform — the unmasked-vs-masked contrast isolates masking as the causal variable.

DL-SCA Setup: Divide-and-Conquer

One CNN profile per S-box column: each of the 64 columns carries 2 unknown key bits $\Rightarrow$ 4 hypotheses per column. Train on 5,000 random-key traces (80/20 split), HW labels 0-5. Random keys are essential: at a fixed key the column label is a function of two public nonce bits (see Gates, trap 3).

Result 1: Every Column Leaks

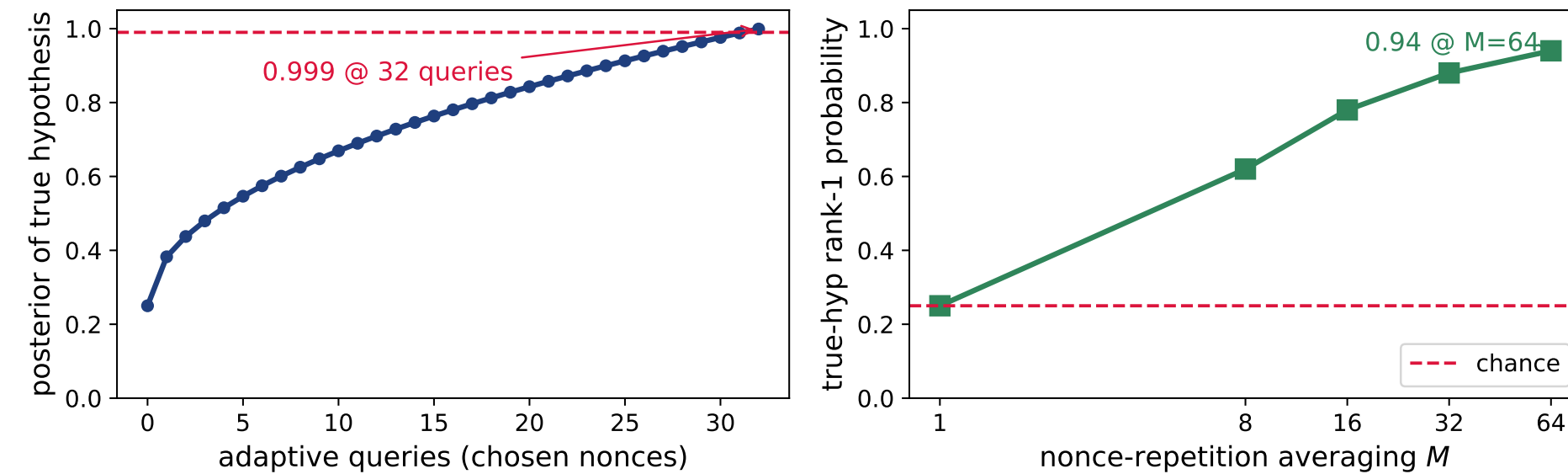


- Held-out random-key traces: key-rank top-1 42.7% mean vs 25% chance; all 64 columns above chance, none below 30%. First-order, real, and per-column.
- Naive single-trace top-1 assembly reaches 37/128 bits — accumulation (below), not single traces, is the recovery path.

Result 2: The Profiling-Domain Gap

Offline profiles transfer poorly to live board sessions — unfine-tuned columns score at chance on-board. Live fine-tuning fixes it: 300 board traces at a known key raise training accuracy to 90.7% (vs 72% majority floor), and the fine-tuned column recovers correct bits at M=1 while unfine-tuned columns sit at chance.

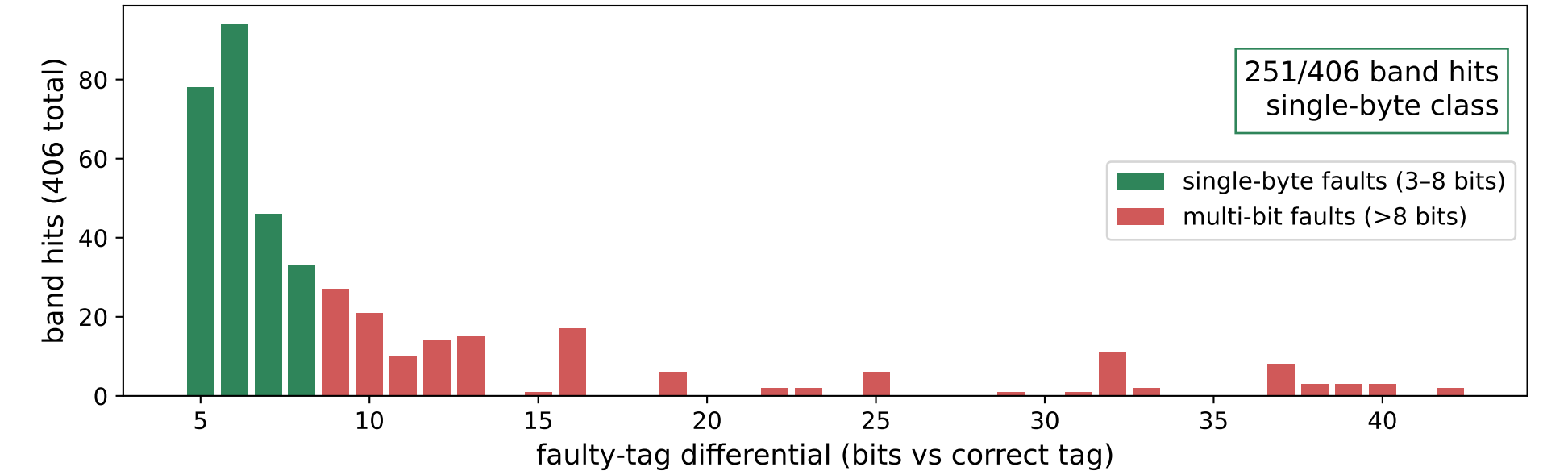
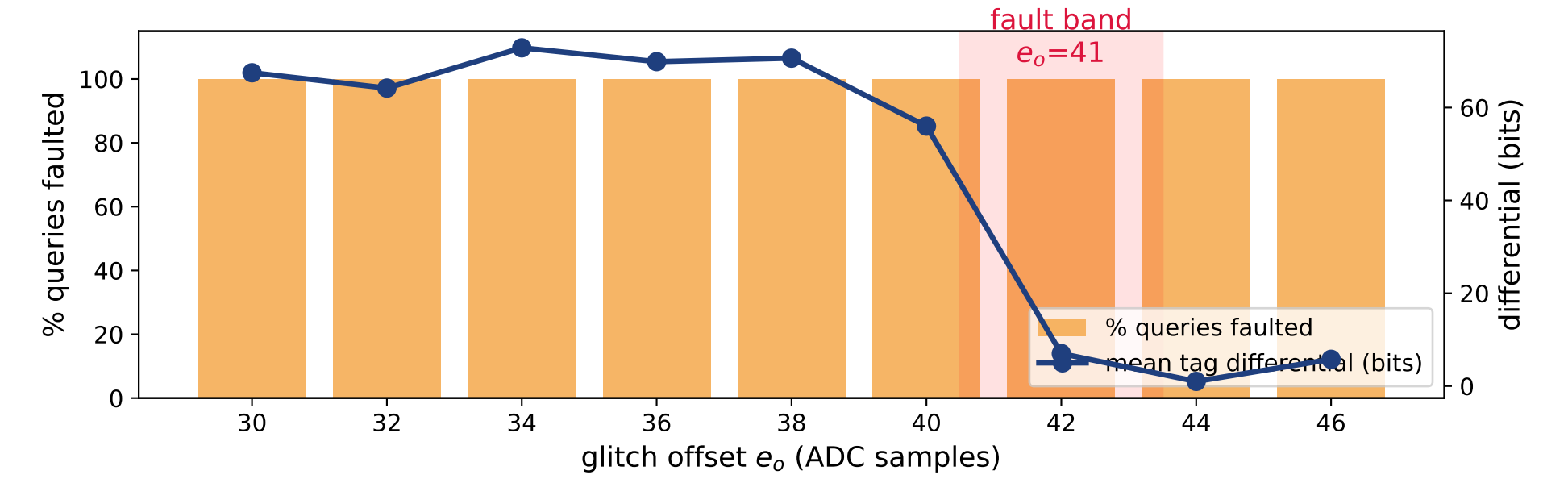
Result 3: Closed-Loop Recovery + SNR Lever



- Posterior-scoring chosen-nonce attack converges to the correct hypothesis in 32 queries (posterior 0.999) — public information only.
- Nonce-repetition averaging: repeating one separating nonce M times lifts true-hyp rank-1 from 0.25 (chance) to 0.94 at M=64; convergence shortens 15 $\rightarrow$ 9 queries/column.
- Full 128-bit assembly: divide-and-conquer over columns + bounded enumeration of least-confident columns, verified by re-encryption. Ongoing work — the pipeline is complete; the remaining gap is on-board fine-tuning budget, not information content.

Take-home: The unmasked NIST core leaks its key bits column-by-column; $\sqrt{M}$ averaging and on-board fine-tuning are the levers that turn weak per-trace signal into confident recovery.

Clock-Glitch Fault Injection



- A stable fault band at $e_o=41$ (width 67): 406 band hits over 153 distinct tags; differentials 27-41 bits.
- 12 clean single-bit round-11 faults localized to exact (column, bit) — 10 distinct state columns — each verified by exact faulty-tag match under simulated single-bit flip.
- Data-selected placement: each nonce faults exactly one column, reproducibly $\Rightarrow$ DFA coverage bounded at one column/query. Nonce-set engineering ($\approx$ 266 nonces, coupon-collector bound) restores coverage.
- Burst glitches: 400/400 faulted but multi-bit (30+ bits) — not composable.

Masked Core (d=1, 2-share)

Same pipeline, 9,151 traces: S-box target at chance for every architecture. First-order masking works; cost $\approx 1.8\times$ throughput, $2.1\times$ LUTs.

Five Leakage-Validation Gates

Artifact trap	Failure mode	Gate
1. Prior-driven “edge”	edge measures the class prior, not leakage	held-out acc > majority floor
2. Prior-only profile + separating-nonce picker	0.993 posterior on the wrong hypothesis	profiles must clear the floor
3. Public-nonce shortcut	model reads nonce-load activity, not key bits	train on random keys only
4. Key-reload bus artifact	$ r \approx 0.6$ fake leakage from key-writes; 64/64 fake columns	survive fixed-key, varying nonces
5. Structured-key alignment	ramp key fakes 75.8% recovery; inverted key scores higher	inversion-invariance + null controls
Traps 4-5 are, to our knowledge, unreported for LWC hardware evaluation.		

References

- [1] NIST, SP 800-232: ASCON Lightweight Cryptography Standards, 2023.
- [2] S. Picek et al., “SCARL,” J. Cryptographic Engineering, 2021.
- [3] Rezaeaezade et al., “One for All, All for Ascon,” ACNS, 2024 (ePrint 2023/1922).